

Tony RuiKang OuYang

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EDUCATION

University of Cambridge

DPhil in Advanced Machine Learning

Cambridge, UK
Oct 2025 – June 2028

- Supervisor [Prof. José Miguel Hernández-Lobato](#) 🌐
- **Fully-funded** by EPSRC DLA Studentship Award

University of Cambridge

MPhil in Machine Learning and Machine Intelligence

Cambridge, UK
Sep 2023 – Aug 2024

- Graduate with **Distinction (Grade: 77%, Top 20%)**
- Course Advisor & Thesis Supervisor: [Prof. José Miguel Hernández-Lobato](#) 🌐
- Thesis: [Energy-based Neural Sampler for Boltzmann Distribution](#) (Grade: 77.5%)

University of Oxford

Visiting Student in Mathematics and Statistics

Oxford, UK
Jan 2022 – Jul 2022

- Supervisor: [Prof. Gesine Reinert](#) 🌐 and [Prof. Geoff Nicholls](#) 🌐
- Fully funded by Harbin Institute of Technology, Shenzhen

Harbin Institute of Technology, shenzhen

BEng in Data Science

Shenzhen, CN
Sep 2019 – Jun 2023

- Graduate with **Distinction (Grade: 90%, Top 10%)**
- Thesis Supervisor: [Yi Zhao](#) 🌐
- Thesis: [Implicit Local Node Embedding Synchronisation for Graph Autoencoders](#) (Grade: 90%)

EXPERIENCES

MolSS reading group

Lead Organiser

London, UK
May. 2025 - present

- A [community](#) 🌐 for machine learning in molecular science

MLG@Cambridge

Research Assistant in Probabilistic Machine Learning

Cambridge, UK
Oct 2024 – Oct 2025

Supervisor: [Prof. José Miguel Hernández-Lobato](#) 🌐

The Alan Turing Institute

Research Assistant

London, UK
Apr. 2022 - Jul. 2022

Supervisor: [Prof. Gesine Reinert](#) 🌐

- Research about scalable Graph Neural Network and Federated Learning for data privacy

PUBLICATIONS

*Equal Contributions

- Yu, Hanlin, **OuYang**, **RuiKang**, Kaushik, Partha, Klami, Arto, Gutmann, Michael U., and Chehab, Omar (2026). [Learning Energy-Based Models from Stochastic Interpolants using Spatiotemporal Differences](#). *Preprint*.
- **OuYang**, **RuiKang***, Grenioux, Louis*, and Hernández-Lobato, José Miguel (2026). [A Diffusive Classification Loss for Learning Energy-based Generative Models](#). *ICML*.
- Yu, James Boran, **OuYang**, **RuiKang**, Horwood, Julien, and Hernández-Lobato, José Miguel (2026). [Stochastic Interpolants in Hilbert Spaces](#). *Preprint*.
- **OuYang**, **RuiKang***, Rissanen, Severi*, He, Jiajun, Chen, Wenlin, Heinonen, Markus, Solin, Arno, and Hernández-Lobato, José Miguel (2025). [Progressive Tempering Sampler with Diffusion](#). *ICML*.
- He, Jiajun*, Du, Yuanqi*, Vargas, Francisco, Zhang, Dinghui, Padhy, Shreyas, **OuYang**, **RuiKang**, Gomes, Carla, and Hernández-Lobato, José Miguel (2025). [No Trick, No Treat: Pursuits and Challenges Towards Simulation-free Training of Neural Samplers](#). *FPI@ICLR*.

- **OuYang, RuiKang***, Qiang, Bo*, and Hernández-Lobato, José Miguel (2025). [BNEM: A Boltzmann Sampler Based on Bootstrapped Noised Energy Matching](#). *TMLR*.
- **OuYang, RuiKang** (2024). [Energy-Based Diffusion Neural Sampler for Boltzmann Densities](#). *MPhil Thesis*.
- **OuYang, RuiKang**, Elliott, Andrew, Limnios, Stratis, Cucuringu, Mihai, and Reinert, Gesine (2023). [L2G2G: A Scalable Local-to-Global Network Embedding with Graph Autoencoders](#). *CNA*.

ACADEMIC SERVICES

- **conference reviewer**: ICLR (2025–2026), ICML (2026), NeurIPS (2025–2026)
 - **award**: silver reviewer @ ICML (2026)
- **workshop reviewer**: [DELTA@ICLR](#) (2025–2026), [FPI@NeurIPS](#) (2025), [SPIGM@NeurIPS](#) (2025), [DynaFront@NeurIPS](#) (2025), [SPIGM@ICML](#) (2026), [FoGen@ICML](#) (2026)
- **organizer**: [MolSS reading group](#) (Lead organizer)

SUPERVISION AND TEACHING

MEng Project Supervision

Sunaabh Trivedi *Better Diffusion Models for Molecular Modelling*

Luran Wang *Towards better Neural Samplers for Boltzmann Distribution*

University of Cambridge

Oct 2024 - June 2025

Oct 2024 - June 2025

MPhil Thesis Supervision

Antonio Franca *Transferable Neural Transports for Parallel Tempering*

James Yu *Stochastic Interpolants in Hilbert Spaces*

University of Cambridge

Apr 2025 - Aug 2025

Apr 2025 - Aug 2025

AWARDS AND ACHIEVEMENTS

Distinction in MPhil Degree, University of Cambridge

Ranked top 20% in MPhil in Machine Learning and Machine Intelligence

2024

Travel and Research Grants, Wolfson College, University of Cambridge

Award for supporting my travelling for conference

2023

Outstanding Graduates, Harbin Institute of Technology (Shenzhen)

Ranked top 10% in HITsz

2023

Distinction in BEng Degree, Harbin Institute of Technology (Shenzhen)

Ranked top 10% in BEng in Data Science

2023

International Exchange Program Scholarship, HITsz

Scholarship for visiting in Oxford, valued at about 200,000 CNY

2022

OTHERS

Athletic

Player, Wolfson College (University of Cambridge) Football Varsity

2023 - 2024

Player, Harbin Institute of Technology (Shenzhen) Football Varsity

2020 - 2023

Captain, Harbin Institute of Technology (Shenzhen) Science School Football Team

2021 - 2023

2nd place of HITsz Football League Season 22-23

2023

2nd place of HITsz Football League Season 20-21

2021

2nd place of HITsz Futsal League Season 20-21

2021

3rd place of HITsz Football League Season 19-20

2020

Languages

- Cantonese (1st Native)
- Chinese Mandarin (2nd Native)
- English (Professional working proficiency; IELTS 7.0)
- French (Elementary proficiency)